1. What is simulation?

   Simulation is the imitation of a real-world process over time using a model.
2. What is a simulation model?

   A simplified representation of a system used to study its behavior.
3. Difference between deterministic and stochastic simulation?

   Deterministic has no randomness; stochastic includes random variables.
4. What is Monte Carlo simulation?

   Uses random sampling to approximate probabilistic outcomes of a system.
5. What is discrete-event simulation?

   Simulation where the system state changes at discrete points in time.
6. What is continuous simulation?

   Simulation where system state changes continuously over time.
7. Difference between static and dynamic simulation?

   Static models do not change with time; dynamic models evolve over time.
8. What is system state in simulation?

   A set of variables describing the system at a specific time.
9. What is an event in simulation?

   An occurrence that changes the system state at a particular time.
10. What is the main purpose of simulation?

    To analyze complex systems and predict behavior without real experiments.
11. What is validation in simulation?

    Ensuring the model accurately represents the real system.
12. What is verification in simulation?

    Checking that the model is correctly implemented according to the design.
13. What is input modeling?

    Process of determining probability distributions for simulation inputs.
14. What is output analysis?

    Studying simulation results to make decisions or estimate performance measures.
15. What is a random variable?

    A variable whose values are determined by chance.

16. Difference between discrete and continuous random variables?

    Discrete has countable outcomes; continuous has infinite possible values.

17. What is probability distribution?

    A function describing the likelihood of different outcomes of a random variable.

18. Difference between uniform and normal distribution?

    Uniform: equal probability; Normal: bell-shaped distribution.

19. What is exponential distribution used for?

    Modeling inter-arrival times in Poisson processes.

20. What is Poisson distribution used for?

    Modeling the number of events in a fixed interval of time or space.

21. What is the role of a seed in random number generation?

    Initializes the random number generator to produce reproducible sequences.

22. Difference between pseudo-random and true random numbers?

    Pseudo-random is algorithmically generated; true random is physically random.

23. What is the inversion method?

    Technique to generate random variables from uniform random numbers using inverse CDF.

24. What is simulation clock?

    Keeps track of simulated time in a model.

25. What is event scheduling approach?

    Simulation method where events are scheduled and executed in chronological order.

26.

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27. Medium Concepts (26–80)

28. What is next-event time advance?

    Simulation time jumps directly to the next scheduled event.

29. What is time-driven simulation?

    Simulation progresses in fixed time increments regardless of events.

30. Difference between time-driven and event-driven simulation?

    Time-driven uses fixed steps; event-driven uses events to advance time.

31. What is state transition diagram?

    Diagram showing possible states and events that cause transitions.

32. What is queuing system in simulation?

    A model representing waiting lines with customers and servers.

33. What are the main components of a queuing system?

    Arrival process, service process, queue discipline, and number of servers.

34. What is arrival rate (λ)?

    Average number of arrivals per unit time in a queuing system.

35. What is service rate (μ)?

    Average number of customers a server can serve per unit time.

36. What is Little's Law?

    $L = \lambda W$, relating average number in system (L) to arrival rate and waiting time.

37. Difference between M/M/1 and M/M/c queues?

    M/M/1: single server; M/M/c: multiple servers, exponential arrivals and service.

38. What is FIFO discipline?

    First-In-First-Out queueing rule.

39. What is LIFO discipline?

    Last-In-First-Out queueing rule.

40. What is priority queue?

    Customers served based on priority rather than arrival order.

41. What is balking?

    Customers decide not to join a queue if it is too long.

42. What is reneging?

    Customers leave the queue after waiting too long.

43. What is simulation warm-up period?

    Initial period ignored to reduce bias from empty system startup.

44. What is steady-state in simulation?

    When system performance measures stabilize over time.

45. What is replication in simulation?

    Running the simulation multiple times to account for randomness.

46. What is common random numbers technique?

    Using same random numbers across scenarios to reduce variance in comparison.

47. What is antithetic variates technique?

    Using negatively correlated random variables to reduce simulation variance.

48. What is control variates technique?

    Using known variables to improve accuracy of simulation estimates.

49. Difference between Monte Carlo and discrete-event simulation?

    Monte Carlo: probabilistic outcomes, no system dynamics; DES: models system events.

50. What is inventory simulation?

    Simulating stock levels, replenishment, and shortages over time.

51. What is EOQ model in simulation?

    Economic Order Quantity; minimizes inventory cost using simulation studies.

52. What is continuous review inventory system?

    Stock level is monitored constantly; order placed when it reaches reorder point.

53. What is periodic review inventory system?

    Stock checked at fixed intervals, and orders placed accordingly.

54. What is safety stock?

    Extra inventory kept to prevent stockouts due to demand variability.

55. What is reorder point?

    Inventory level triggering replenishment order.

56. Difference between open-loop and closed-loop simulation?

    Open-loop has no feedback; closed-loop uses system output to adjust inputs.

57. What is time horizon in simulation?

    Total simulated period considered for analysis.

58. What is random number testing?

    Ensuring generated numbers are uniformly distributed and independent.

59. What is Kolmogorov-Smirnov test in simulation?

    Statistical test to check if data follows a specified distribution.

60. What is Chi-square goodness-of-fit test?

    Measures how well observed data matches expected distribution.

61. What is variance reduction in simulation?

    Techniques used to reduce variability of estimates and improve accuracy.

62. What is bootstrapping in simulation?

    Resampling technique to estimate statistics from limited data.

63. What is simulation optimization?

    Using simulation to find the best system configuration or decision.

64. Difference between deterministic and probabilistic simulation output?

    Deterministic output is fixed; probabilistic output varies per run.

65. What is discrete probability distribution?

    Distribution where random variable has countable outcomes.

66. What is continuous probability distribution?

    Distribution where variable can take infinite values in a range.

67. Difference between empirical and theoretical distributions?

    Empirical derived from data; theoretical based on mathematical function.

68. What is arrival process simulation?

    Modeling the timing and frequency of entities entering the system.

69. What is service process simulation?

    Modeling time required to complete service for entities.

70. Difference between batch and single-server simulation?

    Batch processes multiple entities at once; single-server handles one at a time.

71. What is queue discipline simulation?

    Modeling rules for serving entities in the queue.

72. What is Markov chain simulation?

    System transitions between states probabilistically based only on current state.

73. What is absorbing state in Markov chain?

    State that, once entered, cannot be left.

74. Difference between transient and steady-state analysis?

    Transient observes early system behavior; steady-state observes long-run behavior.

75. What is Monte Carlo integration?

    Estimating integrals using random sampling.

76. What is system dynamics simulation?

    Continuous simulation using feedback loops and differential equations.

77. What is event list in DES?
    Ordered list of future events and their scheduled times.
78. What is simulation clock in DES?
    Keeps track of current simulation time.
79. Difference between fixed-time step and next-event time advance?
    Fixed-step increments time regularly; next-event jumps to next event.
80. What is priority queue in DES?
    Event list organized by event times for execution order.
81. What is warm-up period in queuing simulation?
    Initial period excluded to reduce bias from empty system start.
82. Difference between open and closed queuing networks?
    Open: arrivals/departures; Closed: fixed number of customers circulating.
83. What is multi-server queuing simulation?
    Models systems with multiple servers serving entities concurrently.
84. What is tandem queuing network?
    Queueing system where entities pass sequentially through multiple servers.
85. What is open queuing network simulation?
    Customers enter and leave the system freely; arrivals follow a stochastic process.
86. What is closed queuing network simulation?
    Fixed number of customers circulate among queues; no external arrivals.
87. Difference between serial and parallel queues?
    Serial queues are sequential; parallel queues allow simultaneous service options.
88. What is simulation of priority queues?
    Assigns service order based on priority rather than arrival time.
89. What is fork-join network simulation?
    Models entities splitting into multiple tasks processed in parallel then rejoined.
90. What is transient period in simulation?
    Initial phase where system behavior is not yet stable.
91. What is steady-state period in simulation?
    Phase where system performance measures stabilize over time.

92. What is simulation replication?

Repeating simulation runs to account for randomness and obtain statistical accuracy.

93. What is random number seed in simulation?

Initial value to generate reproducible pseudo-random numbers.

94. What is the role of pseudo-random numbers?

Used to model stochastic variables in simulation.

95. What is uniform random number generation?

Produces numbers equally likely over a specified range.

96. What is inverse transform method?

Generates random variables from uniform random numbers using the inverse CDF.

97. What is rejection method?

Generates random variables by rejecting unsuitable samples.

98. What is variance reduction in simulation?

Techniques to reduce output variability and improve precision.

99. Name common variance reduction techniques.

Common random numbers, antithetic variates, control variates, and stratified sampling.

100. What is control variates method?

Uses correlated known variables to improve estimate accuracy.

101. What is antithetic variates method?

Uses negatively correlated variables to reduce variance.

102. What is stratified sampling in simulation?

Divides population into strata and samples from each to reduce variance.

103. What is bootstrapping in simulation?

Resampling technique to estimate system statistics from limited data.

104. What is stochastic activity network simulation?

Models system with activities having random durations and probabilistic outcomes.

105. What is Monte Carlo method in simulation?

Uses random sampling to estimate probabilistic system behavior.

106. Difference between Monte Carlo and discrete-event simulation?

Monte Carlo uses probabilistic sampling; DES models state changes at events.

107. What is simulation of inventory systems?

Models stock levels, replenishments, and shortages over time.

108. What is EOQ simulation?

Evaluates inventory cost and ordering policies using simulation techniques.

109. What is continuous review inventory simulation?

Stock levels monitored constantly; orders triggered at reorder points.

110. What is periodic review inventory simulation?

Inventory checked at fixed intervals; orders placed accordingly.

111. What is safety stock in simulation?

Extra inventory maintained to prevent stockouts.

112. What is lead time in inventory simulation?

Time between placing and receiving an order.

113. What is backorder in inventory simulation?

Customer demand that is temporarily unfulfilled and recorded for later fulfillment.

114. What is service level in inventory simulation?

Probability of meeting customer demand without stockout.

115. What is lost-sales model?

Inventory model where unmet demand is lost rather than backordered.

116. What is queuing network simulation?

Models interconnected queues to analyze complex systems.

117. What is Jackson network?

Open queuing network with Poisson arrivals and exponential service times.

118. What is Gordon-Newell network?

Closed queuing network where total customers remain constant.

119. Difference between Markovian and non-Markovian queues?

Markovian has memoryless properties; non-Markovian depends on past states.

120. What is Kendall notation in queuing systems?

A/B/C notation describing arrival, service, and number of servers.

121. What is M/M/1 queue?

Single server with Poisson arrivals and exponential service.

122. What is M/M/c queue?

Multiple servers with Poisson arrivals and exponential service.

123. What is M/G/1 queue?

Single server, Poisson arrivals, general service time distribution.

124. What is G/G/1 queue?

General arrival and service distributions with single server.

125. What is bottleneck in simulation?

Stage that limits overall system throughput.

126. What is system throughput?

Number of entities processed per unit time in the system.

127. What is utilization in simulation?

Fraction of time a resource is busy serving entities.

128. What is queue length in simulation?

Number of entities waiting for service at a given time.

129. What is waiting time in simulation?

Time an entity spends in queue before service.

130. What is sojourn time?

Total time an entity spends in the system, including service and waiting.

131. What is discrete-event simulation software?

Tools like Arena, Simio, AnyLogic used to model system events.

132. What is Arena simulation?

Discrete-event simulation software used for manufacturing, queuing, and logistics.

133. What is AnyLogic simulation software?

Multi-method simulation tool supporting DES, system dynamics, and agent-based modeling.

134. What is simulation input analysis?

Determining distributions and parameters for model inputs.

135. What is simulation output analysis?

Evaluating system performance measures using collected simulation data.

136. What is simulation-based optimization?

Using simulation to find the best decision variables or system configuration.